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Applied Analytics Project - Week 13 Assignment

Problem

The objective of this project is to predict the outcomes of NCAA March Madness matchups, specifically if Team 1 is predicted to win or lose the game. Team 1 is defined as the lower-numbered seed. This is a binary classification problem with outputs in the form of probabilities where the target variable is game outcome (win/loss), and outcome is interpreted as the probability that Team 1 wins. The motivation for building this model is to aid in data-driven decision making for bracket selection, specifically when predicting game outcomes in the sports betting environment. In addition, this model will provide interpretable insights into key drivers of basketball game outcomes.

Data Ingestion

Three primary datasets were used for this project. The first dataset includes team-level season statistics with many performance metrics. The second dataset is Team ID mapping to standardize naming. The third dataset is historical NCAA March Madness tournament results from 2003 to 2024. These datasets were combined to form a team-season dataset with one row per team per season, and a game-level dataset with one row per tournament matchup. This dataset is optimal for modeling, since the model will rely on team strength and head-to-head matchups for prediction.

Data Preparation

To clean the dataset, we removed columns with more than 30% missing values. We also standardized team identifiers and merged the datasets to align team statistics with game outcomes. We removed the columns with many missing values in an effort to reduce noise and instability. Cleaning the data helps to make the model more reliable and less biased.

Feature Engineering

This model utilized feature differentials, rather than raw team statistics. The differentials are calculated by subtracting Team 2's value from Team 1's value for various statistics. Some other engineered features include various shooting efficiency metrics, such as FG%, 3PT%, FT%, and eFG%, as well as Assist-to-Turnover ratio. By using differentials, we can predict outcomes based on relative strength instead of absolute values. They help to improve the interpretability of the model and mimic real-world decision logic.

Exploratory Data Analysis (EDA)

EDA focused on validating feature usefulness by comparing feature distribution for wins and losses and identifying features with large separation between classes, which are seed differentials for this model. This method for EDA helped validate which features to include in the modeling step.

Data Validation and Preprocessing

Before modeling, the dataset was checked for missing values, duplicate rows, and any feature scaling issues. This validation step helps ensure the dataset is clean and optimized for better performance in modeling.

Data Splitting Strategy

The dataset was split into three different sets: Training, Validation, and Test. The Training set included all data from the 2003-2021 seasons, with the exception of 2020 since there was no tournament in 2020. The Validation set included data for the 2022 season, and the Test set included data from the 2023-2025 seasons. This structure helps to avoid data leakage and simulates real-world forecasting, where future games are predicted using past data only.

Baseline Modeling (Logistic Regression)

The first model was a basic logistic regression model, which was used as a baseline. This model establishes a performance benchmark and helps provide basic interpretability of results. It also validates that the selected features have predictive value.

Model Development (Final Model Selection)

After logistic regression, we created more complex models, such as random forest and gradient boosting. The final selected model was a random forest model. This model was selected because it had strong predictive performance using evaluation metrics, it handles nonlinear relationships, it has robust feature interactions, and it requires minimal feature scaling. After selecting this model, we tuned the hyperparameters and retrained the final model on the training set.

Model Evaluation

The final random forest model was evaluated using the validation set and test set. The below table shows the performance of the final model compared to the baseline logistic regression model using key evaluation metrics, such as accuracy, precision, recall, f1, and rocAuc. From these results, we can see that the random forest model outperformed the logistic regression model.

	accuracy			precision			recall			f1			rocAuc		
split	test	train	val	test	train	val	test	train	val	test	train	val	test	train	val
model															
logreg_baseline	0.703	0.768	0.698	0.779	0.764	0.677	0.741	0.754	0.700	0.759	0.759	0.689	0.814	0.854	0.790
random_forest	0.734	1.000	0.698	0.797	1.000	0.690	0.778	1.000	0.667	0.788	1.000	0.678	0.814	1.000	0.796

Model Explainability

Explainability techniques were used to identify the top predictive features, understand feature importance, and interpret individual predictions. The key insight from the results is that game outcomes are driven by efficiency metrics, turnover control, and relative team strength.

Bias Detection and Mitigation

The model was evaluated for potential bias in features and any correlations with proxy variables. To mitigate bias, we removed unstable or noisy features, avoided leakage-prone variables, and monitored feature influence. Bias can be present in sports models like this one if certain features dominate unfairly or establish unintended patterns, such as if the model relied too heavily on a variable like seed differential, consistently picking the team with the better seed despite performance metrics.

Model Deployment

Deployment for this model includes saving the trained model, defining input feature schema, and preparing the inference pipeline. Through this process, the model can be reused consistently in future seasons for new predictions.

Monitoring and Maintenance

The two types of drift considered were data drift and concept drift. Data drift is when feature distributions change over time and concept drift is when relationships between features and outcomes change. Our plan to monitor these drifts is to track feature distributions, re-evaluate the model performance each season, and periodically retrain the model with new data.

Production Inference

A production example shows an input of two teams' season profiles. The features for each team or transformed into differentials. Then, the output shows the probability of Team 1 winning, with 0.5 being the win threshold. The below table shows the results from a production example. This validates the full pipeline of our model end-to-end.

	season	team1	team2	team1WinProb	predictedWinner	actualWinner
0	2024	Colorado State	Texas	0.620	Colorado State	Texas
1	2023	Pittsburgh	Xavier	0.215	Xavier	Xavier
2	2023	Saint Mary's	VCU	0.690	Saint Mary's	Saint Mary's

Project Summary

Modeling this problem using relative team strength was critical for success. A random forest model proved to be the best performing and most robust model. A realistic time split between seasons is essential for accurate model evaluation. This project utilized the MDLC, from data ingestion to model monitoring, to create a production-ready workflow.